Sanaul Haque

Matteo_Fanfoni_Proposal

Assignment 108

Document Details

Submission ID
trn:oid:::3117:636671792

Submission Date
Oct 9, 2026, 11:14 PM GMT+6

Download Date
Oct 9, 2026, 11:16 PM GMT+6

File Name
Matteo_Fanfoni_Proposal.pdf

File Size
668.2 KB

8 Pages

2,801 Words

15,323 Characters

22% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **13** Not Cited or Quoted 8%
  Matches with neither in-text citation nor quotation marks
- **4** Missing Quotations 1%
  Matches that are still very similar to source material
- **17** Missing Citation 12%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

16% Internet sources
19% Publications
20% Submitted works (Student Papers)

Match Groups

- **13** Not Cited or Quoted 8%
  Matches with neither in-text citation nor quotation marks
- **4** Missing Quotations 1%
  Matches that are still very similar to source material
- **17** Missing Citation 12%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

16% Internet sources
19% Publications
20% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Student papers	Carnegie Mellon University on 2026-09-28 — 3%
2	Internet	www.biorxiv.org — 2%
3	Publication	Axel A. Balboa, Brayam J. Lazares, Dhiago G. Llanos, Mateo P. Muñoz et al. "Comp... — 2%
4	Student papers	University of the Andes on 2025-11-02 — 2%
5	Student papers	RMIT University on 2023-11-03 — 1%
6	Internet	essay.utwente.nl — 1%
7	Student papers	Unizin, LLC on 2025-10-27 — 1%
8	Publication	Chang Soon Tony Hii, Kok Beng Gan, Huay Woon You, Nasharuddin Zainal et al. "F... — 1%
9	Publication	Hye Chang Rhim, Adam S. Tenforde, Monique S. Burton. "Cross-country/track and... — 1%
10	Internet	www.frontiersin.org — 1%

11 Internet
www.bright-journal.org <1%

12 Internet
www.mdpi.com <1%

13 Publication
Chengkai Su, Chaojie Fu, Yanhong Liang, Kaixin Lan, Yongbin Jin, Hongtao Wang, ... <1%

14 Internet
arxiv.org <1%

15 Publication
Adelene S. M. Wong, Ashleigh V. Morrice-West, R. Chris Whitton, Peta L. Hitchens. ... <1%

16 Student papers
Hartpury College on 2024-03-11 <1%

17 Student papers
University College London on 2026-03-16 <1%

18 Publication
Giuseppe Longo, Rosalba Liguori, Luigi Di Benedetto, Gian Domenico Licciardo, Al... <1%

19 Internet
s3.amazonaws.com <1%

20 Student papers
Athlone Institute of Technology on 2025-11-03 <1%

21 Publication
Ana Virginia Ruescas-Nicolau, Enrique José Medina-Ripoll, Eduardo Parrilla Berna... <1%

22 Publication
Angela M. Senevirathna, Andrew J. Pohl, Matthew J. Jordan, W. Brent Edwards, Re... <1%

23 Student papers
Glasgow Caledonian University on 2024-03-25 <1%

24 Internet
par.nsf.gov <1%

25 Internet

www.researchgate.net <1%

IARCO RESEARCH PROPOSAL

Full Legal Name: Matteo Fanfoni

Institution: American School of Milan

Category: Junior

Class/Grade/Year: 12

Country: Italy

Major: Computer Science

Submission Date: September 21, 2026

Registered Email Address: mattefanfoni@gmail.com

Research Topic: Does Smartphone-Video Ground Contact Time Estimation Generalize Beyond a Pilot Cohort? A Proposed Validation Study in Developing Middle-Distance Runners in Milan, Italy

Title: Does Smartphone-Video Ground Contact Time Estimation Generalize Beyond a Pilot Cohort? A Proposed Validation Study in Developing Middle-Distance Runners in Milan, Italy

Research Problem

17 Ground contact time (how long each foot stays on the ground when running), step cadence, stride length, and vertical oscillation are running gait parameters that affect running economy, performance or injury incidence. Higher cadence and smaller vertical displacement drive a lower energy cost of running [1]; ground contact time can discriminate between winners and losers in a championship 1500m final [2]; and 22 in collegiate cross-country runners, step rate was a predictor of bone stress injury, with risk falling about 5% per extra step per minute [3]. Measuring these gait metrics with lab accuracy means marker-based 3D motion capture or an instrumented treadmill, something which is confined to sports-science departments or elite athletes, while developing runners have normally no access to this feedback and show incidence estimates as high as 68% over a single competitive season [4].

A smartphone camera can capture a run, and a deep learning pose-estimation model like BlazePose can derive running gait metrics with near lab-grade accuracy [5]. One such example is OpenCap, which has shown smartphone motion capture working at a fraction of a lab's cost and time, at real scale [6].

As part of an open-source project, a pilot pipeline was tested on five runners at race pace, and its ground-contact-time accuracy against annotated ground truth achieved a pre-registered target of less than 10 milliseconds using two independent estimators (registering 7.57 ms and 3.99 ms mean absolute error, respectively). That result came from a small cohort, and it hasn't been tested yet whether the accuracy holds for a larger group.

Existing Literature

Four areas of research support this proposal.

First, a growing set of studies validates markerless, smartphone-based pose estimation against marker-based motion capture for running gait. Young et al. validated Google's BlazePose against a Vicon reference system for running gait and found agreement with intraclass correlation coefficients above 0.75 for most temporal parameters [5]. Uhlrich et al. proposed an open-source pipeline that computes kinematic parameters from smartphone video and validated it in a 100-subject field study [6]. Preece et al. found that elite and recreational runners move differently [7]. The combined evidence of these studies show that smartphone pipelines can approach lab accuracy, but ability level changes running mechanics and for that reason the pipeline may not necessarily generalize.

Second, studies of elite 800m and 1500m runners during races show which kinematic parameters change with pace, in the distances this research targets. Hanley et al. found that elite 1500m finalists ran faster mainly by raising step frequency rather than step length, and that contact time was a factor in the race result [2]. A related study of the women's 800m final found the same thing: faster running within the race came with higher cadence and shorter contact and flight times [8]. Both point to ground contact time and cadence as drivers of performance and as such parameters worth tracking.

Third, the ground-truth method this proposal reuses, manual frame-by-frame video annotation in Kinovea, has its own validation literature. Puig-Diví et al. found that Kinovea gives reliable coordinate and distance measurements at close distances and sagittal view [9]. Spanos et al. found Kinovea reliable for most gait parameters while identifying the tracking of the pelvis-position as a weaker result [10], something that matters for any Kinovea measurement involving the hip or pelvis like the one foreseen in this research proposal.

Fourth, youth distance runners are documented to be prone to injuries [4], [11]. Cadence has been studied as a target for reducing that risk [12], and a study of collegiate cross-country runners found a lower step rate associated with bone stress injury [3]. That is one parameter the pipeline in this research proposal measures, in a similar population.

Research Question

How far does the ground-contact-time accuracy from a smartphone-video, deep-learning biomechanics pipeline, shown on an initial five-runner pilot cohort against a pre-registered target of 10 ms or less (7.57 ms mean absolute error for a neural network under leave-one-runner-out cross-validation, where each runner is held out from the model that scores them, and 3.99 ms for a physics-based engine evaluated on that same cohort's own tuned parameters), hold up on a larger, more varied cohort of adolescent and young-adult developing middle-distance runners training in Milan, Italy?

Methodology

This study builds on an existing validated pipeline and reuses the pilot cohort's protocol so the two sets of results are comparable. Figure 1 shows what is being measured.

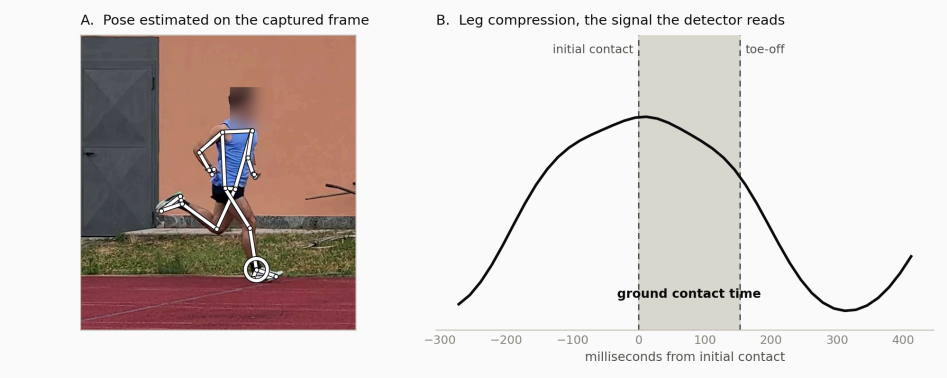


Fig. 1. Measurement variables, illustrated for a pilot video clip. (A) The estimated pose sketched on an image of the first contact frame, with the foot used in the stance phase marked; the head of the individual is blurred. (B) The leg-compression data, read by the contact detector, with the stance phase shaded between the initial contact and toe off. The human observer marks those two frames manually, and the difference between the two is what is measured in this study.

Population.

Participants come from a group of athletes based in the Milan area, Italy, which is the same context where the videos of the pilot cohort were shot. The group includes adolescents and young adults (roughly ages

16 to 20) who train competitively for 800m or 1500m events. The target sample is about 10 to 20 runners, depending on runners' availability, which is two to four times the size of the pilot cohort, and a sample large enough to cover differences in strike pattern, sex, and main event within the defined age group.

Data collection.

Each participant will be filmed at a resolution of 4K and 60 frames per second with a smartphone from a fixed camera distance and sagittal view. Calibration poles will be used to double-check pixel-to-metre conversion (the primary method being the use of runners' own height). This is both the same protocol as the pilot study and the filming condition earlier work found most accurate for frame-based video annotation [9]. Each runner will be filmed at three pre-calculated, target race-paces: threshold, 1500m, and 800m, since earlier work shows the way runners increase speed at these intensities differs from slower paces and, most importantly, is where gait signature (stride vs cadence) becomes visible [2], [8].

Height measurement.

Each participant's height will be measured before the start of the shooting session and recorded next to the height declared by the runners. The pipeline converts pixels to metres partly through a per-runner body-scale estimate, so an error in a runner's declared height carries into every stride length that runner produces. The pilot cohort wasn't measured, leaving per-runner scale differences and declared height error mixed together and impossible to separate afterwards. Recording both numbers takes an error out of the stride-length results, and sizes self-reported height error.

Ground truth annotation.

An independent observer will mark initial contact and toe-off using frame-by-frame scroll mode in Kinovea to derive the ground-truth for ground contact time. This also allows to derive cadence and stride length. A few clips will be annotated again to measure intra-rater repeatability. This is the same annotation method used for the pilot cohort.

Analysis.

The physics-based engine and the neural network built on the pilot cohort will be run on the new cohort's data without retraining, using leave-one-runner-out cross-validation for both estimators. The neural network's figure was already held out by runner, while the engine's wasponce evaluated on the same cohort used to tune its own parameters. A previous test re-derived two of the engine's distributional constants leave-one-runner-out and found the cohort figure barely moved, but two other constants in its contact-detection path still haven't been tested that way. Running the whole engine under leave-one-runner-out cross-validation on this new cohort closes that gap. The main outcome is ground-contact-time mean absolute error against the new ground truth. Each new runner's error will be compared to the pilot cohort's five runner-level errors with a two-sample test built for small, uneven, non-normal samples, such as a Mann-Whitney U test. Recruitment also addresses a known limit of the pilot design: in the original cohort, each runner had one consistent foot-strike pattern, so strike type and runner identity couldn't be separated. This study will deliberately recruit runners with different strike patterns within the same age and training-level group, so strike-pattern effects can be checked on their own. Leave-one-out cross-validation on a small cohort carries its own bias, shown in recent work to skew performance estimates [13].

Ethical considerations.
Written consent will be requested from all adult participants or from parents of minors. All video will be stored securely and used only for the research. No medical claims will be made from the data.

Research Topic

I designed the pipeline to run two independent estimators, a physics-based engine and a neural network. The network takes the runner's declared pace as an input through a feature-wise modulation layer, so one model can adjust to different paces instead of needing a separate model for each one. The project also publishes its own negative results, including a case where a more complex design did not beat a simpler one.

25 This research addresses two problems that compound each other. Running-related overuse injury is common in adolescent and developing distance runners, with incidence as high as 68% in a single competitive season [4], and a lower step rate, one of the parameters this pipeline measures, is prospectively associated with that risk [3]. At the same time, the tools that measure those parameters at lab accuracy are out of reach, financially and logistically, for almost every runner outside an elite or university training setting. A validated, smartphone-only alternative would change who gets this kind of feedback at all, reaching the developing and non-elite runners least likely to be monitored today. That holds only if the measurement works outside the small, controlled conditions it was first shown in. This study asks whether the pilot result is a foundation or an artifact of a small, favorable first sample.

Roadblocks and Potential Limitations

There are four potential problems with this study, and naming them now is cheaper than discovering them later.

Recruitment.
There is an expectation of 10 to 20 runners, contingent on availability. Should the number be at the low end of the scale, the comparison with the pilot's five runners is weakened, and failure to find a result becomes indistinguishable from sampling problems. Videoing each runner at three different speeds generates multiple clips per runner, but the unit of analysis remains the runner, and so clips cannot be used instead.

Strike-pattern balance.
Trying to recruit a mix of strike-patterns requires that such runners show up, and should the recruitment again produce all the same runners, then the confounding of strike type with runner identity will persist.

The reference has a floor of its own.
The human ground truth involves marking the frames manually, which means that 60 frames per second imply 16.7 ms per frame. In this case, the 10 ms accuracy of the reference cannot be tested by the measuring device because it is less accurate, so the write-up will specify that floor instead of quietly going beyond it.

One observer.

One annotator allows for checking intra-rater repeatability, but not inter-rater agreement. An additional annotator who will mark the clips selected randomly would address this problem, and if there is somebody who can do this, he or she will be hired.

One limitation is introduced on purpose. This time the new cohort is trained in the same area of Milan as the pilot study, so the design allows testing generalizability across runners, but not across different venues or different cameras or lighting conditions. This will be said in the write-up.

Post-Program Plan

Once the study is completed, all results will be documented and submitted to student research journals, such as the Youth Research Journal, the Columbia Junior Science Journal, and the International Journal of High School Research. A piece of research will also end up going in the annual congress of the Italian Society of Clinical Movement Analysis.

Publication will not be the only communication channel, as the pipeline is already open source under an MIT license with a public demo wesbite. All other useful documents such as the new cohort's aggregate results, the annotation protocol, and the analysis scripts, will also be released.

References

[1] B. Van Hooren, I. Jukic, M. Cox, K. G. Frenken, I. Bautista, and I. S. Moore, "The relationship between running biomechanics and running economy: a systematic review and meta-analysis of observational studies," *Sports Med.*, vol. 54, no. 5, pp. 1269-1316, May 2024, doi: 10.1007/s40279-024-01997-3. Accessed: Sep. 14, 2026. [Online]. Available: https://doi.org/10.1007/s40279-024-01997-3

[2] B. Hanley, A. Bissas, S. Merlino, and G. T. Burns, "Changes in running biomechanics during the 2017 IAAF World Championships men's 1500 m final," *Scand. J. Med. Sci. Sports*, vol. 33, no. 6, pp. 931-942, 2023, doi: 10.1111/sms.14331. Accessed: Sep. 13, 2026. [Online]. Available: https://doi.org/10.1111/sms.14331

[3] S. A. Kliethermes, M. R. Stiffler-Joachim, C. M. Wille, J. L. Sanfilippo, P. Zavala, and B. C. Heiderscheit, "Lower step rate is associated with a higher risk of bone stress injury: a prospective study of collegiate cross country runners," *Br. J. Sports Med.*, vol. 55, no. 15, pp. 851-856, Aug. 2021, doi: 10.1136/bjsports-2020-103833. Accessed: Sep. 14, 2026. [Online]. Available: https://doi.org/10.1136/bjsports-2020-103833

[4] R. H. Mann, C. D. McKay, B. C. Clift, C. A. Williams, and A. R. Barker, "Injuries and Training Practices in Competitive Adolescent Distance Runners: A Retrospective Cross-Sectional Study," *Front. Sports Act. Living*, vol. 3, art. 664632, 2021, doi: 10.3389/fspor.2021.664632. Accessed: Sep. 13, 2026. [Online]. Available: https://doi.org/10.3389/fspor.2021.664632

[5] F. Young, R. Mason, R. Morris, S. Stuart, and A. Godfrey, "Internet-of-Things-Enabled Markerless Running Gait Assessment from a Single Smartphone Camera," *Sensors*, vol. 23, no. 2, art. 696, Jan. 2023, doi: 10.3390/s23020696. Accessed: Sep. 13, 2026. [Online]. Available: https://doi.org/10.3390/s23020696

[6] S. D. Uhlrich, A. Falisse, L. Kidziński, et al., "OpenCap: Human movement dynamics from smartphone videos," *PLOS Comput. Biol.*, vol. 19, no. 10, art. e1011462, 2023, doi: 10.1371/journal.pcbi.1011462. Accessed: Sep. 13, 2026. [Online]. Available: https://doi.org/10.1371/journal.pcbi.1011462

[7] S. J. Preece, C. Bramah, and D. Mason, "The biomechanical characteristics of high-performance endurance running," *Eur. J. Sport Sci.*, vol. 19, no. 6, pp. 784-792, 2019, doi: 10.1080/17461391.2018.1554707. Accessed: Sep. 13, 2026. [Online]. Available: https://doi.org/10.1080/17461391.2018.1554707

[8] B. Hanley, S. Merlino, and A. Bissas, "Biomechanics of World-Class 800 m Women at the 2017 IAAF World Championships," *Front. Sports Act. Living*, vol. 4, art. 834813, 2022, doi: 10.3389/fspor.2022.834813. Accessed: Sep. 13, 2026. [Online]. Available: https://doi.org/10.3389/fspor.2022.834813

[9] A. Puig-Diví, C. Escalona-Marfil, J. M. Padullés-Riu, A. Busquets, X. Padullés-Chando, and D. Marcos-Ruiz, "Validity and reliability of the Kinovea program in obtaining angles and distances using coordinates in 4 perspectives," *PLoS ONE*, vol. 14, no. 6, art. e0216448, 2019, doi:

10.1371/journal.pone.0216448. Accessed: Sep. 13, 2026. [Online]. Available: https://doi.org/10.1371/journal.pone.0216448

[10] S. Spanos, A. Kanellopoulos, K. Petropoulakos, Z. Dimitriadis, I. Siasios, and I. Poulis, "Reliability and applicability of a low-cost, camera-based gait evaluation method for clinical use," *Expert Rev. Med. Devices*, vol. 20, no. 1, pp. 63-70, 2023, doi: 10.1080/17434440.2023.2171289. Accessed: Sep. 13, 2026. [Online]. Available: https://doi.org/10.1080/17434440.2023.2171289

[11] T. Paz, R. N. Meyers, C. N. Faverio, Y. Wang, E. M. Vosburg, and D. J. Clewley, "Youth Distance Running and Lower Extremity Injury: A Systematic Review," *Int. J. Environ. Res. Public Health*, vol. 18, no. 14, art. 7542, Jul. 2021, doi: 10.3390/ijerph18147542. Accessed: Sep. 13, 2026. [Online]. Available: https://doi.org/10.3390/ijerph18147542

[12] I. Figueiredo, M. Reis e Silva, and J. E. Sousa, "The Influence of Running Cadence on Biomechanics and Injury Prevention: A Systematic Review," *Cureus*, vol. 17, no. 8, art. e90322, Aug. 2025, doi: 10.7759/cureus.90322. Accessed: Sep. 13, 2026. [Online]. Available: https://doi.org/10.7759/cureus.90322

[13] G. I. Austin, I. Pe'er, and T. Korem, "Distributional bias compromises leave-one-out cross-validation," *Sci. Adv.*, vol. 11, no. 48, art. eadx6976, 2025, doi: 10.1126/sciadv.adx6976. Accessed: Sep. 13, 2026. [Online]. Available: https://doi.org/10.1126/sciadv.adx6976